$$\left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} & \frac{-1}{6} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & \frac{1}{5} & 1 & \frac{-1}{5} \end{array}\right] \begin{array}{l} A_6 = \\ B_6 = \\ C_6 = \end{array}$$